PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Detail
Pentester Name (Cybersecurity Professional)	Fortunate Dube
Program/Batch	Networkwalks Cybersecurity Internship (Batch084 2026)
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com (secured written permission already) 2. My own local LAN Network (192.168.56.0/24)
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (**W2-PM1**) and scanning my own local network with Zenmap (**W2-PM5**). One module covers the footprinting phase and the other covers the scanning phase, so together they demonstrate how an ethical hacker or security auditor moves from gathering public reconnaissance data to mapping live hosts on a network. It is the Week 2 project deliverable of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (for footprinting) and with Zenmap installed (for active network scanning). Every step below includes the exact command used, the result observed, a screenshot reference as evidence, and a note on why the finding matters from an attacker's and defender's point of view.

3. Tools Used

The table below lists each tool used in this report and its security purpose:

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, registrar, name servers).
whatweb	Fingerprint web technologies (server banner, CMS, plugins, IP address).
nslookup	Resolve the domain name to its routable IPv4 address using DNS.
curl -I	Read HTTP response headers and uncover hidden endpoints/caching headers.
wafw00f	Detect whether a Web Application Firewall (WAF) protects the website.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV, and BIND daemon version).

Zenmap (Nmap GUI)	Scan the local subnet to discover live hosts, IP addresses, MAC addresses, and generate network topology.
Windows CMD	Identify local IP address, subnet mask, and network adapter MAC address.

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon. Each tool was used to collect a distinct layer of intelligence about the target:

1. **WHOIS:** Queried public registry databases using whois networkwalks.com. The results revealed domain registration through GoDaddy and identified authoritative nameservers pointing to HostGator (NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM).
2. **WhatWeb:** Fingerprinted the website's tech stack using whatweb networkwalks.com. The scan revealed an Apache web server, WordPress CMS, and the WP Download Manager plugin, alongside front-end frameworks (Bootstrap, jQuery).
3. **Nslookup:** Resolved the domain name to its direct IPv4 address using nslookup networkwalks.com. The domain resolved to 192.232.216.135.
4. **Curl -I:** Inspected the HTTP response headers using curl -I https://networkwalks.com. The headers revealed Server: Apache, endurance caching levels, and exposed the WordPress REST API endpoint at /wp-json/.
5. **Wafw00f:** Fingerprinted the active firewall protection using wafw00f networkwalks.com. The tool confirmed that the website is protected by **ModSecurity (SpiderLabs) WAF**.
6. **DNSRecon:** Enumerated complete DNS zone records using dnsrecon -d networkwalks.com. The scan mapped the mail server (mail.networkwalks.com), SPF anti-spoofing policies (v=spf1 ...), cPanel autodiscovery service records (SRV), and identified the BIND DNS daemon version (9.16.23-RH).

4.2 Network Scanning with Zenmap (W2-PM5)

For the second activity, I used Zenmap (GUI frontend for Nmap) to perform host discovery and network mapping within my authorized local subnet:

1. **Subnet & IP Identification:** I ran ipconfig in the Windows Command Prompt. My local network interface had an IP address of 192.168.56.1 with a subnet mask of 255.255.255.0, defining a local LAN subnet of 192.168.56.0/24.
2. **Host Discovery (Ping Scan):** In Zenmap, I configured the target as 192.168.56.0/24 with the **Ping scan** profile (nmap -sn 192.168.56.0/24). Nmap scanned all 256 IP addresses in 13.71 seconds and confirmed:
    a. **Live Host Discovered:** 192.168.56.1 (Host is up)
    b. **MAC Address:** Obtained via ipconfig /all under the network adapter.
3. **Topology Mapping:** I switched to the **Topology** tab in Zenmap, opened the **Legend**, and exported the visual graphic in PDF format as topography.pdf.

5. Risk Analysis / Impact

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology & plugin exposure	WhatWeb identified WordPress and WP Download Manager	Attackers may query CVE vulnerability databases for known exploits affecting these versions	● Medium
2	Server IP address directly identifiable	Nslookup resolved domain to `192.232.216.135`	Allows adversaries to scan the origin server directly or bypass front-end protections	● Low
3	HTTP response headers leak REST API	Curl returned `Server: Apache` and `/wp-json/` link	May assist technology fingerprinting and unauthenticated user enumeration	● Low

4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Knowing the exact WAF brand allows attackers to craft evasion and encoding payloads	● Low
5	DNS version and infrastructure exposure	DNSRecon exposed BIND software version (`9.16.23-RH`) and SRV records	Disclosing DNS software versions allows targeting of daemon-specific vulnerabilities	● Medium
6	Active hosts and topology identifiable	Zenmap ping sweep mapped active IP and subnet	In unmanaged networks, visible active hosts can expose unprotected endpoints	● Low

Risk Level Key: ● *Critical* | ● *Medium* | ● *Low*

Note: The items listed above are observations from passive footprinting and authorized host discovery, not confirmed vulnerabilities. No active exploitation was conducted.

6. Recommendations & Remediation

1. **Suppress Server & Software Banners:** Configure Apache (ServerTokens Prod, ServerSignature Off) and BIND DNS daemons to suppress software version disclosure in HTTP headers and DNS queries.
2. **Restrict WordPress REST API Access:** Disable unauthenticated REST API enumeration for user endpoints (/wp-json/wp/v2/users) to prevent reconnaissance of valid usernames.
3. **Maintain WAF Signatures:** Ensure ModSecurity OWASP Core Rule Set (CRS) is regularly updated to defend against emerging zero-day exploit patterns.
4. **Enforce Strict SPF/DMARC Policies:** Regularly audit DNS TXT and SPF records to prevent domain spoofing and phishing attacks.

5. **Periodic Network Scanning:** Regularly perform internal discovery scans with Nmap/Zenmap to identify unauthorized or rogue devices connected to the corporate LAN.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I successfully completed practical modules covering both **Footprinting & Reconnaissance** and **Active Network Scanning**.

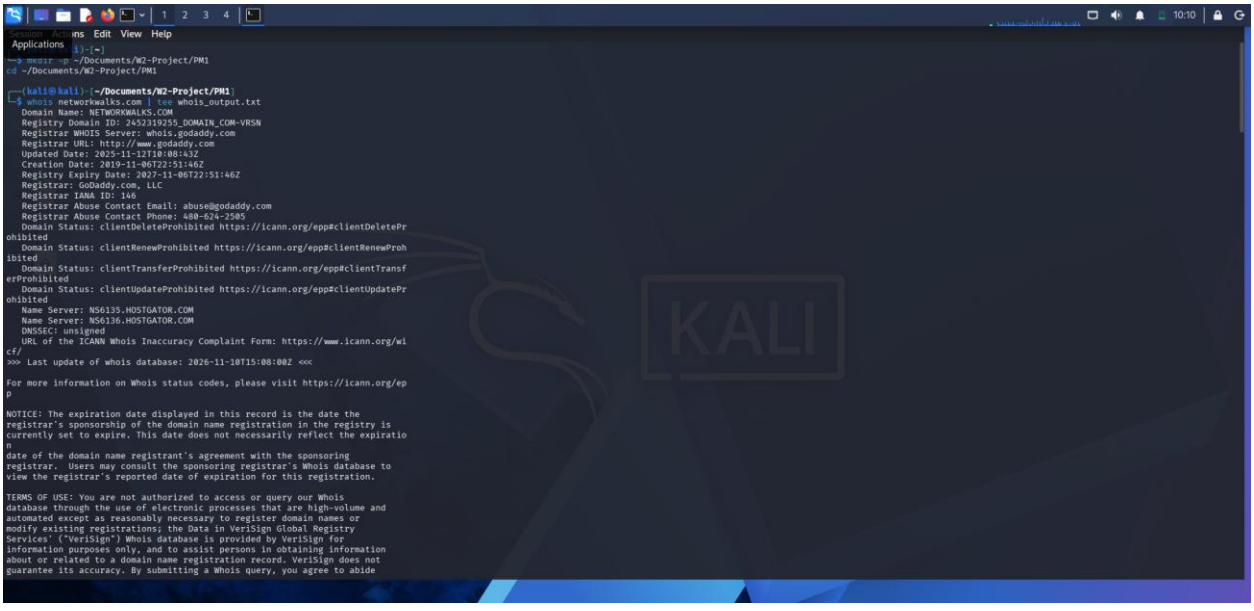
In the footprinting module (**W2-PM1**), I gained hands-on experience using six fundamental Kali Linux tools to map out a public organization's digital footprint without launching intrusive attacks. In the network scanning module (**W2-PM5**), I applied CIDR subnetting principles, conducted host discovery using Zenmap, and analyzed network topology.

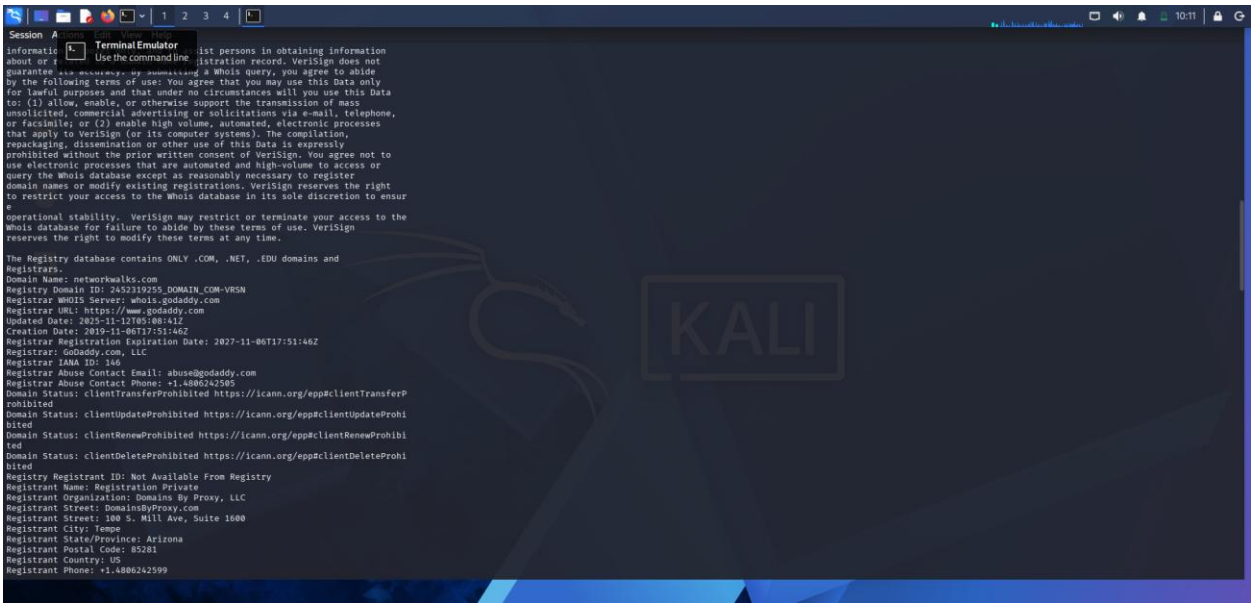
These practical exercises demonstrated that thorough information gathering is the bedrock of cybersecurity defense. By auditing what our infrastructure leaks to the public, security engineers can proactively harden systems before malicious actors exploit that knowledge.

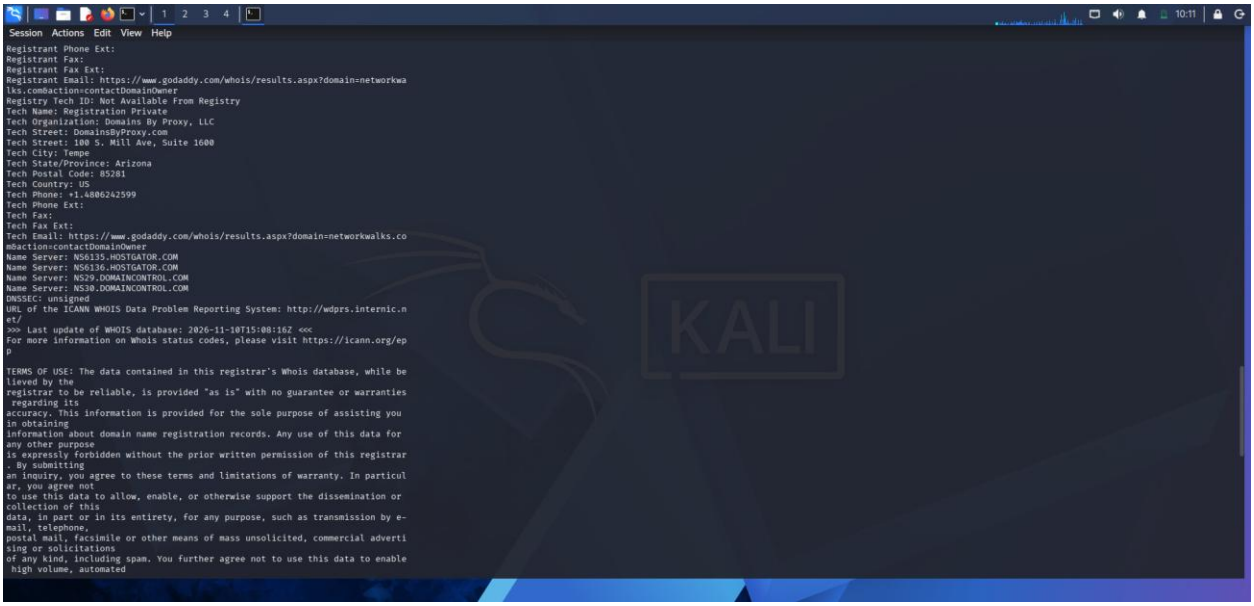
8. Evidences Collected

All screenshots captured during the activities are stored in the screenshots/ directory:

WHOIS query output showing registrar and nameservers.




Session Actions Edit View Help
Terminal Emulator
Use the command line
10:11
informatic ist persons in obtaining information
about or r istration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensur
e
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferP
rohibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhi
bited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibi
ted
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhi
bited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599

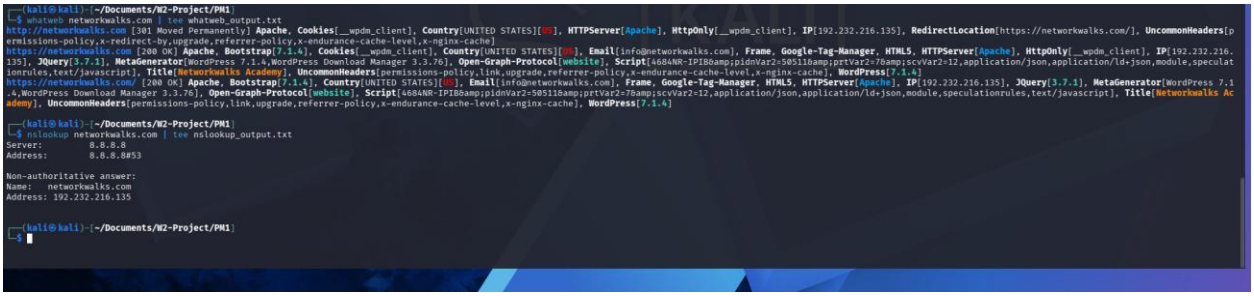

Session Actions Edit View Help
10:11
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwa
lks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.co
m&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM
Name Server: NS30.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.n
et/
>>> Last update of WHOIS database: 2026-11-10T15:08:16Z <<<
For more information on Whois status codes, please visit https://icann.org/ep
p
TERMS OF USE: The data contained in this registrar's Whois database, while be
lieved by the
registrar to be reliable, is provided "as is" with no guarantee or warranties
regarding its
accuracy. This information is provided for the sole purpose of assisting you
in obtaining
information about domain name registration records. Any use of this data for
any other purpose
is expressly forbidden without the prior written permission of this registrar
. By submitting
an inquiry, you agree to these terms and limitations of warranty. In particul
ar, you agree not
to use this data to allow, enable, or otherwise support the dissemination or
collection of this
data, in part or in its entirety, for any purpose, such as transmission by e-
mail, telephone,
postal mail, facsimile or other means of mass unsolicited, commercial adverti
sing or solicitations
of any kind, including spam. You further agree not to use this data to enable
high volume, automated



WhatWeb output showing WordPress, plugins, and web stack.



nslookup query resolving domain to IPv4 address.



Curl-l HTTP response headers showing server banners and REST API links.

```
Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135


┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$ curl -I https://networkwalks.com | tee curl_output.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:--  0:00:03 --:--:--     0
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https:/
/www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=55a49c3ebcc03c30286d37bef3987b2; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 15:42:41 GMT
server: Apache


┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$
```

Wafw00f detection of ModSecurity (SpiderLabs) WAF.

```
┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$ wafw00f networkwalks.com | tee wafw00f_output.txt

                ,____
               (  W00f! )
                `  ___/
                 ,,
404 Hack Not Found
                                  405 Not Allowed
                                403 Forbidden
502 Bad Gateway                 500 Internal Error

                    ~ WAFW00F : v2.4.2 ~
      The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$
```

dnsrecon Complete DNS enumeration records and BIND version.

```
[*] Number of requests: 2

┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$ dnsrecon -d networkwalks.com | tee dnsrecon_output.txt
2026-11-10T10:52:45.071971-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-10T10:52:45.072717-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-10T10:52:46.821660-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-10T10:52:47.972013-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-10T10:52:48.982304-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-10T10:52:49.133788-0500 ERROR    Recursion enabled on NS Server 50.87.144.87
2026-11-10T10:52:49.159596-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-10T10:52:49.237443-0500 ERROR    Recursion enabled on NS Server 192.232.216.131
2026-11-10T10:52:53.735767-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-10T10:52:54.275695-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-10T10:52:55.673692-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-10T10:52:55.674169-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-10T10:52:57.283558-0500 INFO Enumerating SRV Records
2026-11-10T10:53:17.577965-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-10T10:53:17.578271-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-10T10:53:17.578424-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-10T10:53:17.578552-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-10T10:53:17.578670-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-10T10:53:17.578787-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-10T10:53:17.579097-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-10T10:53:17.579349-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-10T10:53:17.581351-0500 INFO 8 Records Found
2026-11-10T10:53:17.582528-0500 INFO Completed enumeration for domain: networkwalks.com


┌──(kali㉿kali)-[~/Documents/W2-Project/PM1]
└─$
```